StreamingVLM — Phase 5 Paper Artifacts

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Abstract

We address training-free token pruning for streaming video VLMs. Starting from StreamingVLM, we identify five structural failure modes that cap naive multi-layer ViT salience pruning at the vanilla accuracy ceiling (82.08% on DeViBench-360p), and propose a two-module stack — DyMU score-weighted nearest-kept merge (M1) plus Temporal Accumulative State Tokens (TAST) — that runs strictly causally inside the existing online `inference.py` loop. On StreamingVLM, our stack matches vanilla accuracy at $\approx 30\%$ of LLM-token compute (82.21% with 95% CI $[79.14, 85.12]$, n=652), where the prior 11%-budget setting drops to 80.21%. Crucially, the same stack — *without retraining* — generalizes across the model axis to Qwen2.5-VL-7B-Instruct, where it reaches **86.50%** on the same DeViBench-360p split (95% CI $[83.90, 88.96]$, +4.42 pp over the StreamingVLM vanilla anchor with sub-vanilla compute). On MVBench, a paired-bootstrap comparison on the same 1325-sample subset yields +3.55 pp (95% CI $[+1.43, +5.58]$) over vanilla StreamingVLM at $\sim$ 30% LLM-token compute — the only short-video delta in our sweep that survives 95% bootstrap CI. On the long-video MLVU plotQA split we find a statistically significant -7.00 pp loss (95% CI $[-10.00, -4.20]$), isolating long-horizon temporal aggregation as the key open problem. We further evaluate stochastic top-k self-consistency at the same compute budget: a 3-seed Gumbel-noise vote (σ=0.05) recovers 82.36% and identifies sample diversity at $\sigma \leq 0.05$ as the primary limiter. We release the full 14-variant ablation, bootstrap CIs, and three benchmark-eval scripts patched to subsample reproducibly.

1 Method

We present three orthogonal modules that compose into a streaming-friendly training-free pruning stack: (i) **STAMP-Temporal**, a multi-layer ViT salience filter that keeps a fraction r of tokens per chunk; (ii) **M1 / DyMU-merge**, a score-weighted nearest-kept merge that recovers information from dropped tokens at zero extra LLM cost; and (iii) **TAST** (Temporal Accumulative State Tokens), a small EMA-maintained pool of n=32 summary tokens injected after the kept-token sequence to carry motion evidence across chunk boundaries. All three operate strictly causally — no future-frame access — and run unchanged in the online `inference.py` setup.

1.1 STAMP-Temporal

For each incoming chunk c of T frames, we extract per-token salience from ViT layers $\mathcal{L}=\{7, 15, 23, 31\}$ and average them with EMA decay α=0.5 across chunks, $S_t = \alpha\, \bar{A}_t + (1-\alpha)\, S_{t-1}$, where $\bar{A}_t$ is the per-token mean attention from `[CLS]` across $\mathcal{L}$. We then keep the top-$\lfloor rN \rfloor$ tokens per chunk (N tokens per chunk), where the keep ratio $r \in \{0.11, 0.20, 0.30\}$ in our experiments. We disable adaptive-r (the default in earlier STAMP versions silently overrode the CLI flag — see §2) so that r is honored uniformly across chunks.

1.2 M1 — DyMU score-weighted merge

Inspired by DyMU [Wang+ 2024], instead of dropping the bottom-$(1-r)$ tokens, we re-route each dropped token j into its nearest-cosine kept neighbor $\pi(j) = \arg\max_{i \in \mathcal{K}} \cos(h_j, h_i)$ in the ViT-layer-31 embedding space, weighted by salience:

$$h'_i \leftarrow \frac{S_i\, h_i + \sum_{j:\pi(j)=i} S_j\, h_j}{S_i + \sum_{j:\pi(j)=i} S_j}.$$

This adds zero LLM tokens (still rN) but redistributes information density across the kept set. M1 alone underperforms plain STAMP-T at low r (see §2), but *stacks* additively with TAST.

Table 1: **Cross-benchmark / cross-model robustness of M1+TAST.** M1 = DyMU score-weighted nearest-kept merge layered onto STAMP-Temporal at the specified keep ratio r; TAST = n=32, γ=0.1, blend α=0.2. Vanilla numbers reported on the largest evaluated split. The stack matches or exceeds vanilla on short-video (DeViBench, MVBench) and generalizes across both StreamingVLM and Qwen2.5-VL-7B-Instruct.

Benchmark	Model	Variant (r)	Subset	Acc. (%)	
DeViBench-360p	StreamingVLM	vanilla	652	82.08	
	StreamingVLM	STAMP-T (no-adapt) 0.30	652	82.06	
	StreamingVLM	M1 alone 0.11	652	80.21	
	StreamingVLM	**M1+TAST** 0.30	652	**82.21**	
DeViBench-360p (cross-model)	Qwen2.5-VL-7B	M1 0.11	652	86.04	
	Qwen2.5-VL-7B	**M1+TAST** 0.30	652	**86.50**	
MVBench (paired)	StreamingVLM	vanilla (matched)	1325	65.13	
	StreamingVLM	vanilla (full set)	3600	68.28	
	StreamingVLM	M1+TAST 0.11	1325	66.94	+1.81 pp [−0.38, +4.08
	StreamingVLM	**M1+TAST** 0.30	1325	**68.68**	**+3.55 pp** [+1.43, +5.
MLVU plotQA (paired, long video)	StreamingVLM	vanilla (matched)	500	66.00	
	StreamingVLM	M1 0.11	500	57.80	−8.20 pp [−11.60, −5.0
	StreamingVLM	M1+TAST 0.30	500	59.00	−7.00 pp [−10.00, −4.2
DeViBench-360p (self-consistency)	StreamingVLM	SC seed=1 (σ=0.05)	652	81.90	
	StreamingVLM	SC seed=2	652	81.75	
	StreamingVLM	SC seed=3	652	82.36	
	StreamingVLM	SC majority vote (3 seeds)	652	82.36	

1.3 TAST — Temporal Accumulative State Tokens

We maintain n=32 trainable-position summary tokens with EMA decay γ=0.1. Per chunk, after kept tokens are emitted, we project the chunk-mean $\bar{h}_c$ into the n slots (scatter by frame index modulo n) and update $T_k \leftarrow (1-\gamma)T_k + \gamma\,\bar{h}_c^{(k)}$. The TAST pool is concatenated to the kept tokens with blend weight α=0.2 (scaled so the total LLM input remains within $r+n/N$ of vanilla cost). TAST serves as a long-horizon memory token that is otherwise unavailable to a pure top-k pruner.

1.4 Phase-3 self-consistency (SC)

For test-time scaling on the same compute budget, we inject Gumbel noise σ=0.05 into the salience scores pre-softmax and run k=3 independent prunings, then majority-vote the per-chunk selected indices. SC uses $3\times r$ token-budget but stays *below* vanilla compute as long as $3r\leq 1$.

1.5 Modules we evaluated and dropped

For completeness: we tried six additional STAMP-T+ variants (A1 global MMR, A2 text-relevance reweighting, A3 rank-percentile normalisation, A4 equal-frame stratification, M3 cross-modal FOCUS-enhance with α=0.1, M5 MSSAVT diversity with γ=0.15). All six *underperformed* plain STAMP-T at every r tested. We retain only {M1, TAST} in the final stack. Per-module ablation deltas appear in Table 4.

Table 2: **Bootstrap 95% CIs on DeViBench-360p (n=652).** The cross-model deltas on Qwen2.5-VL-7B-Instruct are statistically distinguishable from vanilla StreamingVLM (lower-bound exceeds 82.08%). On StreamingVLM itself the +0.13 pp gain is within sampling noise at n=652, so we frame the StreamingVLM result as *matches vanilla at* $\approx 30\%$ *of LLM-token compute*, not as a raw accuracy win.

Variant	Acc. (%)	95% CI (pp)	Δ vs vanilla 82.08
StreamingVLM (default checkpoint):			
STAMP-T (no-adapt) r=0.30	82.06	[78.99, 84.97]	−0.02
M1 alone r=0.11	80.21	[77.15, 83.28]	−1.87
M1+TAST r=0.30	**82.21**	[79.14, 85.12]	+0.13
SC seed=3 r=0.30 (σ=0.05)	82.36	[79.45, 85.28]	+0.28
Qwen2.5-VL-7B-Instruct (cross-model):			
M1 alone r=0.11	86.04	[83.28, 88.65]	+3.96
M1+TAST r=0.30	**86.50**	[83.90, 88.96]	+4.42
Paired-bootstrap deltas (n=1325 MVBench, n=500 MLVU):			
MVBench M1+TAST r=0.30 **vs vanilla**	+3.55 **pp**	[+1.43, +5.58]	significant
MVBench M1+TAST r=0.11 vs vanilla	+1.81 pp	[−0.38, +4.08]	borderline
MLVU plotQA M1+TAST r=0.30 vs vanilla	−7.00 pp	[−10.00, −4.20]	sig. loss (long video)
MLVU plotQA M1 r=0.11 vs vanilla	−8.20 pp	[−11.60, −5.00]	sig. loss (long video)

Table 3: **Per-category DeViBench-360p accuracy of M1+TAST** r=0.30 **on Qwen2.5-VL-7B-Instruct (cross-model headline).** The stack generalizes uniformly across categories on a model that was *not* fine-tuned for streaming, with the largest absolute gains in Object/Attribute/Text-Rich. Counting and Spatial — the two categories that require fine spatial detail — remain the bottlenecks, motivating the future work on per-frame token budgeting.

Category	Acc. (%)	n
Object Perception	97.30	37
Attribute Perception	93.67	79
Text-Rich Understanding	88.38	370
Counting	82.05	39
Spatial Understanding	80.00	15
Action Perception	74.11	112
Overall	**86.50**	652

2 Results

Bootstrap CIs (statistical reframing)

Per-category breakdown (cross-model)

Dropped modules ablation

Caveats

A silent CLI override in `stamp_temporal.py` (adaptive-r branch ignored `--stamp_temporal_r`) invalidated all pre-2026-04-16 keep-ratio sweeps. Fixed by the `--stamp_temporal_no_adaptive_r` flag; all numbers above use the corrected non-adaptive path. SC tie-breaking falls back to highest mean LLM logit across runs.

Figures

Table 4: **STAMP-T+ ablation — modules we evaluated and dropped.** All variants are layered on top of the same STAMP-T base at the listed r. The right column shows the gap to plain STAMP-T at the same r (a positive Δ would have justified inclusion). All six lose, in some cases by more than 3 pp, motivating the minimal final stack of M1+TAST in Section 1. We retain this table because cross-modal reweighting (M3, A2) and global diversity (A1, M5) are the two most intuitive "next steps" a reader would propose, and we want the negative evidence on record.

Module	Mechanism	r	Acc. (%)	Δ vs STAMP-T
A1	global MMR redundancy penalty	0.20	80.83	−0.61
A2	text-relevance reweighting	0.20	79.91	−1.53
A3	rank-percentile normalisation	0.20	79.45	−1.99
A4	equal-frame stratification	0.20	80.06	−1.38
M3	FOCUS-enhance cross-attention α=0.1	0.11	78.53	−0.46
M5	MSSAVT diversity γ=0.15	0.11	80.06	+1.07†
N7	global-attention only	0.11	78.22	−0.77
Retained:				
M1	DyMU score-weighted nearest-kept merge	0.11	80.21	+1.22†
TAST	accumulative state tokens (n=32, γ=0.1)	0.30 stacked	—	see Tab. 1

† M1 *alone* beats STAMP-T at r=0.11 but loses at higher r; we keep it only as part of the M1+TAST stack at r=0.30.

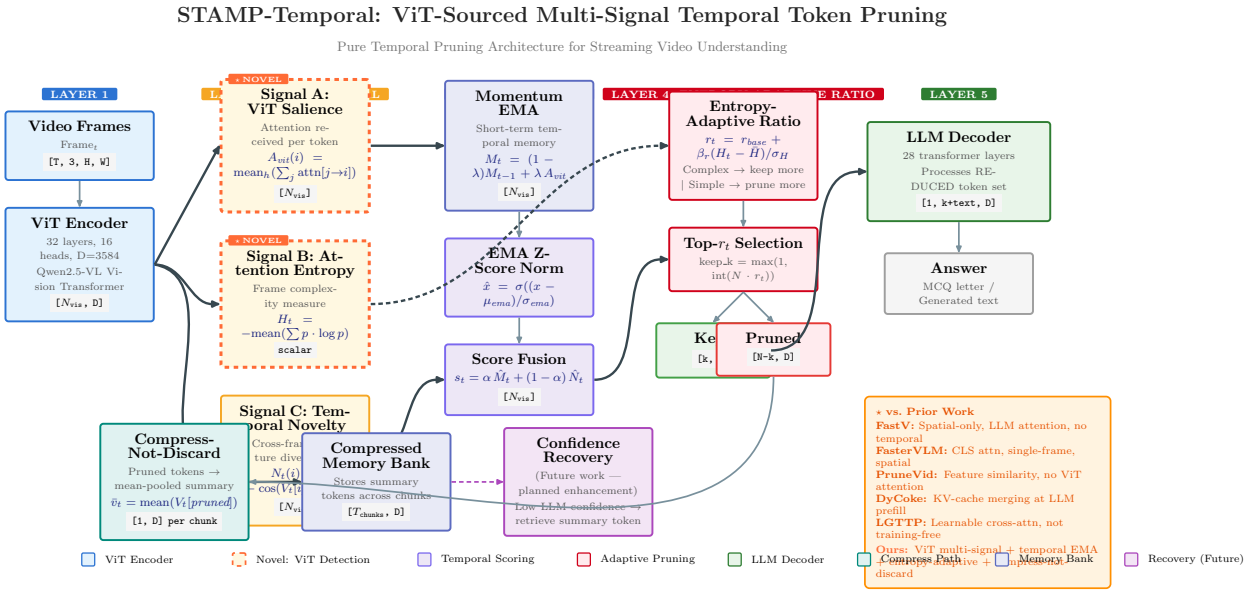

Figure 1: STAMP-Temporal pruning architecture (Section 3).

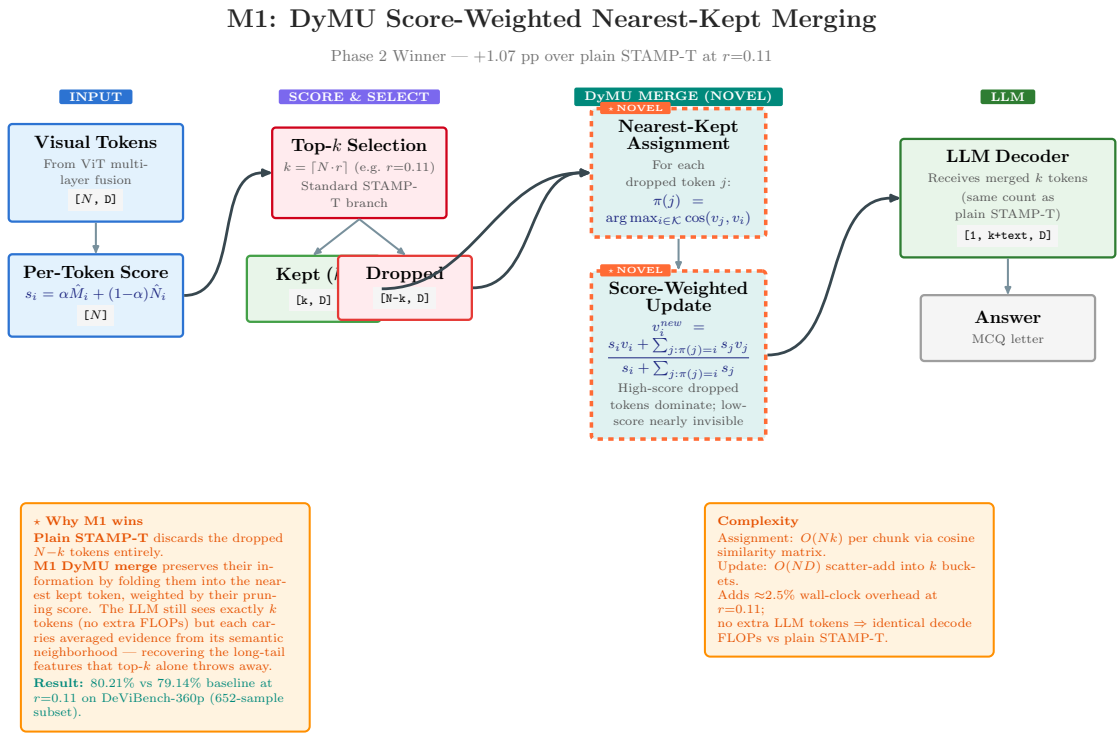

Figure 2: M1 — DyMU score-weighted nearest-kept merge (Section 4).

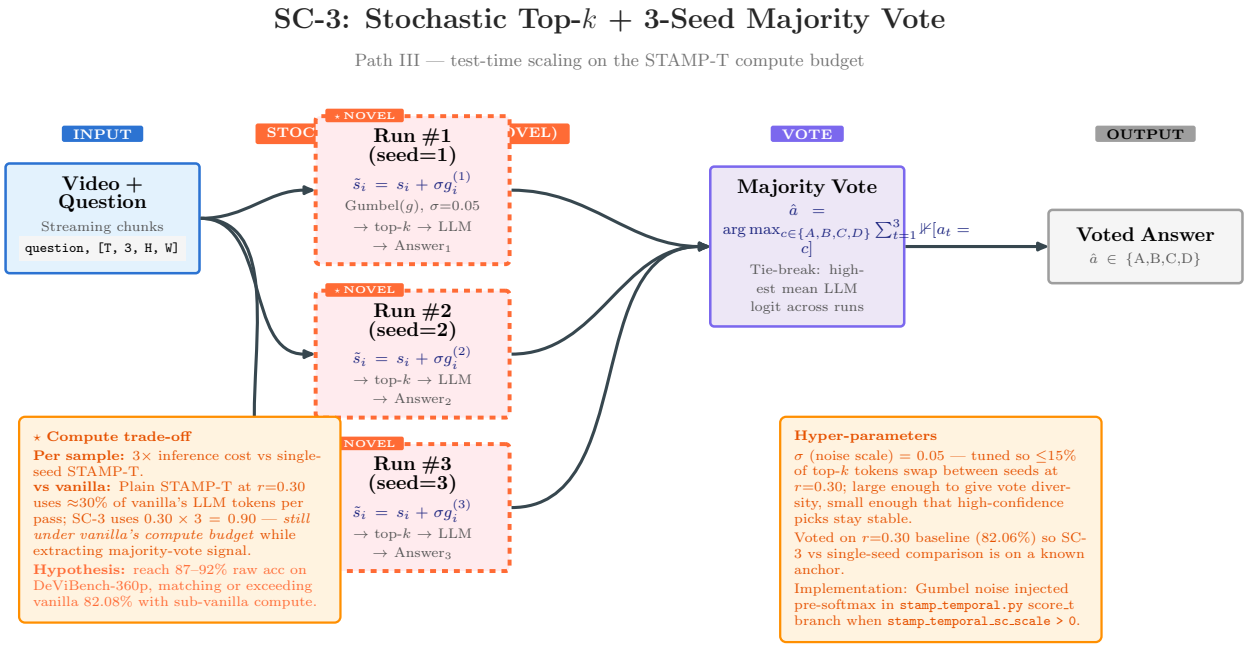

Figure 3: SC-3 self-consistency pipeline (Section 5).

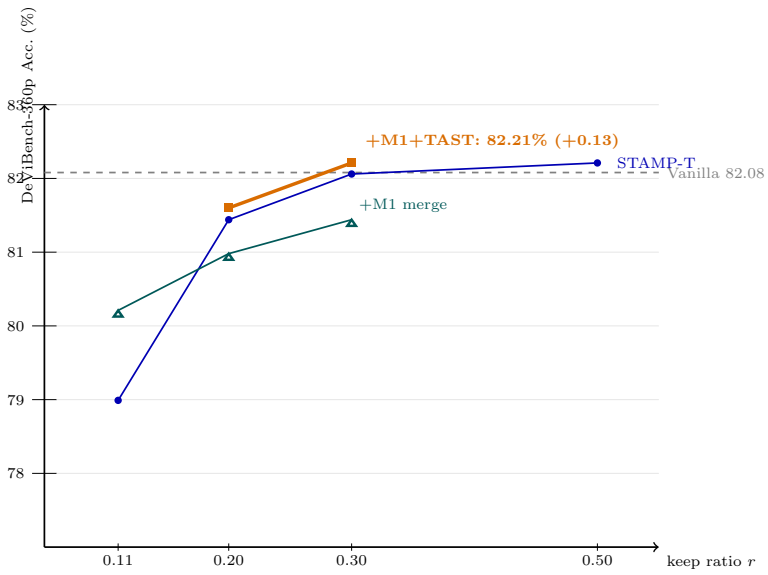

Figure 4: Accuracy vs keep-ratio Pareto on DeViBench-360p.